

RESEARCH

Quantum-Inspired Hybrid Machine Learning Framework for Scalable Computational Next-Generation Technologies

K. Balasubramanian · T. Ganesan · K. Kalaivani · S. Hemamalini

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Abstract

This paper examines quantum computing's revolutionary potential to overcome classical computing's limitations and unleash unprecedented computational power to promote innovation across many industries. Quantum computing might revolutionize AI, cryptography, simulations, and finance by solving scalability, optimization, and real-time processing issues. Traditional computer approaches struggle with complicated difficulties, especially in machine learning (ML), which demands more efficient processing due to enormous datasets and numerous sophisticated models. We introduced Quantum-Inspired Hybrid Machine Learning Models (Q-IHMLM) that combine traditional and quantum techniques. This paradigm aims to bridge this gap. Q-IHMLM uses quantum mechanics to boost computational efficiency, which speeds up data processing, improves forecasts, and improves financial modeling for portfolio optimization, risk assessment, and fraud detection. The suggested technique is adaptable since it is employed in big data analytics, AI model training, optimization, and financial analysis. The abstract should be refined to explicitly highlight the quantum circuit structure used, detailing gate composition and qubit connectivity. It must also describe the encoding mechanisms that map classical data into quantum states. Additionally, the role of algorithmic differentiation in training and optimization should be clarified. These elements will provide a more complete technical foundation for the proposed work. Q-IHMLM speeds up problem-solving while maintaining accuracy and saving processing resources, making real-world applications more efficient and scalable. We found that Q-IHMLM outperforms other approaches in real-time data processing, accuracy, and scalability. This achievement affects the banking industry and other organizations and sets the way for computer technology improvements.

Keywords Big data analytics · Computational breakthroughs · Hybrid models · Machine learning · Optimization · Quantum computing

1 Introduction

Quantum computing, based on quantum physics, is a new alternative to computer science. It might alter practically all enterprises worldwide [1]. Modern technology's computing needs, especially for simulations that require a lot of processing, AI, and cryptography [2]. New computing paradigms must be fast, scalable, and precise to



handle the growing complexity of computational problems and data [3]. New quantum computing technology is expected to outperform traditional computers [4]. This is because the technology is new. Quantum computing's most basic version can solve computationally difficult issues [5]. The quantum bit's superposition ability is one of its most essential properties [6].

The real-world application of quantum computing remains a significant challenge due to the complexity of quantum algorithms and the limited development of quantum hardware [7]. Although classical ML excels in pattern recognition, deep learning (DL), and predictive analysis applications, its reliance on older computer systems is a limitation [8]. Therefore, researchers are considering integrating advanced machine-learning methods with quantum concepts [9]. The new hybrid models optimize data processes, increase prediction accuracy, and enhance scalability by utilizing advanced quantum-inspired techniques [10]. The computation-intensive tasks of traditional ML are achieved using concepts of quantum entanglement and superposition in the new model [11]. This allows faster model training on larger datasets.

These functions enable the maximization of integration and intelligence empirically, delivering increased speed and accuracy, and tackling complex computational tasks [12]. This is essential for addressing the growing demand for efficient and adaptable computational systems [13]. The paper's results demonstrate how the capability of AI can be strengthened by quantum computing, which enables the evolution of technology in the coming years [14].

1.1 Motivation

Consequently, these new paradigms in computing address the exponential increase in data consumption and the growing complexity within computer operations [15]. Concerning optimization, cryptography, and AI, traditional systems are limited in terms of speed, accuracy, and scalability [16]. These challenges have been well addressed by quantum computers combined with ML in parallel processing [17].

1.2 Statement of the Problem

Modern classical computing systems struggle to optimize and scale for high-complexity applications like artificial intelligence, optimization, and big data analytics. Traditional systems struggle to handle increased data volumes in terms of speed, accuracy, and resource use. In the case of ML models, it is necessary to utilize leading-edge computational structures that optimize resources, increase performance, and scale very well without sacrificing accuracy.

1.3 Contribution of this Paper

- This paper outlines the development of a hybrid method that combines quantum algorithms and traditional ML approaches to achieve enhanced computational efficiency while maintaining accuracy.
- The method shows great flexibility and can be used in several other fields, such as AI model training, big data analytics, and optimization, specifically focusing on finance applications such as risk assessment and fraud detection.
- The analytical evaluation conducted in this paper demonstrates the strengths of Q-IHMLM over the current solutions by showing realistic model accuracy, time processing, and comprehensive data scalability.

The remainder of this paper is structured as follows: In Sect. 2, the related work of quantum computing is studied. In Sect. 3, the proposed methodology of Q-IHMLM is explained. In Sect. 4, the efficiency of Q-IHMLM is discussed and analyzed. Finally, in Sect. 5, the paper concludes with a discussion of future work.

2 Related Work

It explores the transformative possibilities of quantum computing throughout several fields, including power systems, economics, high-energy physics, and future warfare. Stressing both opportunities and constraints for the future, research on the potential of quantum technologies to address challenging issues like scalability, energy efficiency, and computing concerns shows how these developments will revolutionize companies and industries.

2.1 Quantum Systems Exploration

This work suggests studying numerous quantum systems to solve important problems, including error sensitivity, decoherence, and scalability. Overcoming technological limitations and advancing quantum computing toward practical applications depends on middleware, quantum hardware development, and interdisciplinary research [18]. This method aims to accelerate field development by revolutionizing various areas through innovative ideas and strategic investments.

2.2 Quantum-FinTech Integration

In response to portfolio optimization, risk management, fraud detection, and secure encryption, this paper examines how quantum computing can impact the FinTech industry. It emphasizes how Generation Z's search for moral, effective, and focused solutions increases acceptance [19]. Recognizing technological immaturity, legal limitations, and adoption challenges, the study examines quantum applications in distributed finance (DeFi), blockchain, and real-time analytics. The report provides strategic insights and highlights how quantum computing can revolutionize the finance industry.

2.3 Quantum Solutions for Power Systems

Driven by the integration of renewable energy, grid operation, and real-time monitoring, this work analyzes quantum computing to address the high processing demands in contemporary power systems [20]. It examines how quantum algorithms might address computing challenges in sectors such as dynamic security analysis, stochastic power flow calculations, and large-scale data analytics. The paper presents known quantum solutions, analyzes the computational loads of classical and quantum computing for power system applications, and proposes future development pathways to expand quantum capabilities.

2.4 Energy-Efficient Quantum Systems

Highlighting quantum computing, communication, and sensing, this look examines the energy efficiency of several quantum systems. It compares numerous quantum technologies to determine the most promising techniques for subsequent developments[21]. Moreover, the paper demonstrates how nanotechnology can enhance the efficiency of quantum systems, thereby supporting ecological stability and the energy economy. The outcomes deepen the discussion on energy-efficient quantum technology, providing academics and practitioners with valuable insights.

2.5 Quantum Financing System

This paper proposes a theoretical basis for a quantum finance system, including circuits, quantum financing algorithms, and various financial scenarios[22]. It highlights how the more accurate, secure, and agile performance of quantum computers can transform the financial markets. The research addresses the challenges and future

directions of quantum finance from both scientific and operational perspectives, providing an informative analysis for academics, professionals, and policymakers.

2.6 Quantum Transformation Framework

This paper, with basic ideas and consequences for sectors including finance, healthcare, and logistics, explores the development of quantum computing[23]. It highlights opportunities, including Quantum-as-a-Service (QaaS) and enhanced AI capabilities, while also addressing ethical concerns and challenges. Emphasizing the need for joint efforts among enterprises, politicians, and engineers to utilize quantum technologies responsibly, the essay provides a comprehensive overview of the evolving potential and complexity of the quantum age.

2.7 Quantum High-Energy Physics Applications

It explores the possibilities of quantum computing to transform high-energy physics by addressing demanding computational problems that are either difficult or impossible for current methodologies. We review the current state of quantum computing in the field and benchmark applications in theoretical and experimental physics, including those related to the Large Hadron Collider [24]. Additionally, the paper presents resource estimates for potential applications, highlighting the importance of continued research to establish use cases for near-term quantum computers.

2.8 Emerging Technologies in Future Warfare

With scientific advances in fields such as AI, quantum computing, cyber warfare, and space, this anthology attempts to explore the evolving nature of future warfare. It explores the challenges and issues arising from these advances, such as the changes in the conduct of military operations, strategy, and tactics [25]. Table 1 shows the comparison of existing methods.

Soumen Pal et al. [26] suggested Quantum Computing in the Next-Generation Computational Biology Landscape. Within the context of next-generation computational biology, this study examines the current state of quantum computing technology and its potential applications in addressing challenges in molecular biology. Data storage capacity utilizing quantum gates, the operation of quantum systems storing information as qubits, and the fundamental idea of quantum computing were all laid forth in the article. Second, the review covered the fundamentals of quantum computing, including q-hardware, q-processors, and q-annealing. In the same breath, the piece covered quantum algorithms, such as the Grover search algorithm, discrete algorithms, and factorization techniques. In the context of a Quantum-Inspired Hybrid Machine Learning Framework for Scalable Computational Next-Generation Technologies, the relationship to prior work should be articulated by emphasizing how existing quantum-inspired models have primarily addressed isolated aspects such as feature encoding, dimensionality reduction, or optimization, but often lacked scalability and integration with classical machine learning pipelines. The proposed approach builds on these foundations by combining quantum-inspired optimization strategies with classical learning mechanisms, thereby overcoming computational bottlenecks and enabling scalability for real-world applications. By explicitly contrasting these contributions with the limitations of earlier studies, the framework highlights its novelty in bridging quantum principles with practical, next-generation computational technologies.

Sebastian Anetey Shamo [19] proposed the Generational Perspective on Gen Z's Influence on Adoption and Growth based on Quantum Computing in the Evolution of Fintech. From enhancing the efficiency of decentralized finance (DeFi) and blockchain to providing real-time analytics and fraud prevention, the study examines the potential applications of quantum computing in FinTech. It highlights how the demands of Gen Z for efficiency, transparency, and ethics are shaping market dynamics. Furthermore, the research notes that technology

Table 1 The comparison of existing methods

Reference number	Methods	Environment	Advantages	Performance	Limitations
[18]	Quantum Systems Exploration	Quantum Middleware	Overcomes technological limitations to facilitate practical use	High processing speed	High error sensitivity, decoherence issues, and scalability concerns
[19]	Quantum-FinTech Integration	Blockchain	Enables portfolio optimization, fraud detection, and real-time analytics	Medium efficiency	Technological immaturity, legal limits, and adoption challenges
[20]	Quantum Solutions for Power Systems	Smart Grid	Tackles high processing demands in power systems; enhances grid operation	High computational efficiency	Limited quantum capabilities need further research for scalability
[21]	Energy-Efficient Quantum Systems	Nanotechnology	Improves energy efficiency and ecological stability	High energy efficiency	High computational cost; limited quantum hardware capabilities
[22]	Quantum Financing System	Quantum Circuits	Transforms financial systems with improved agility and security	Highly accurate predictions	Incomplete understanding of regulatory and operational challenges
[23]	Quantum Transformation Framework	Cloud Computing	Explores QaaS enhanced AI capabilities; addresses ethical concerns	Enhanced AI capabilities	Ethical and practical implementation hurdles
[24]	Quantum High-Energy Physics Apps	Energy Physics	Solves complex computational problems in physics; aids LHC research	Superior computational power	Resource constraints and limited near-term applications
[25]	Emerging Technologies in Future Warfare	Autonomous Systems	Addresses evolving warfare tactics with AI, quantum, and cyber tools	High strategic impact	Ethical dilemmas, attribution challenges, and potential for misuse

immaturity, high expenses, regulatory gaps, and knowledge limitations are obstacles to adoption, and it suggests strategic solutions to overcome these problems.

The articles shed light on the applications of quantum technology across various domains, from enhancing power system energy efficiency to transforming banking through quantum algorithms. Disciplines such as quantum finance, high-energy physics, and next-generation warfare are examined, with an emphasis on the shifting capabilities of quantum computers. The pieces cover technological capabilities and limitations, offering perspectives for further research and development.

3 Proposed Method

This paper explores the intersection of quantum computing and ML, intending to enhance IoT-edge-cloud management, quantum circuit design, and financial systems. It demands hybrid models, security systems for real-time financial applications, iterative procedures, quantum-integrated circuits, and effective resource management. Stressing the importance of quantum-inspired solutions, the discussion promotes computer security, scalability, and efficiency in various disciplines, including finance, industry, and healthcare. The

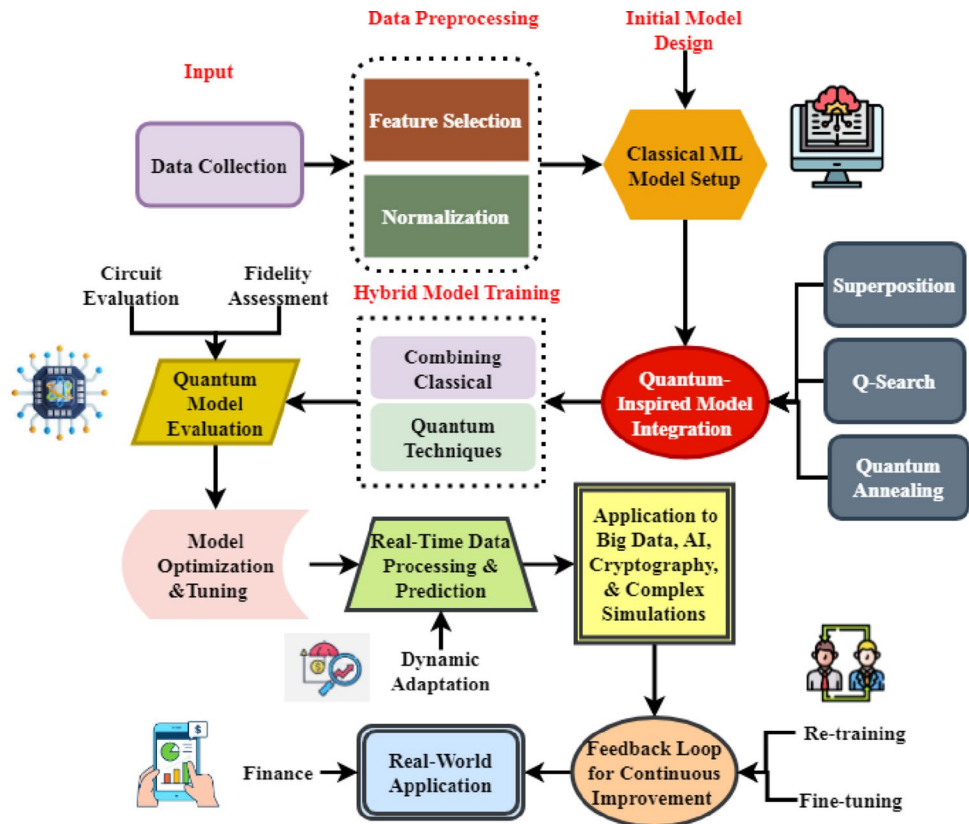
proposed Q-IHMLM architecture demonstrates potential by integrating QAOA and Grover-inspired modules, yet the manuscript would benefit from a clearer representation of the complete data flow. Specifically, it should outline how raw input undergoes classical preprocessing (e.g., normalization, feature extraction), followed by the encoding of processed data into quantum states for QAOA optimization and Grover-inspired search operations. Furthermore, the mechanism of translating quantum outputs back into classical representations for final prediction or decision-making should be explicitly detailed. Such a depiction would provide readers with a holistic understanding of the end-to-end pipeline, ensuring both technical clarity and practical applicability. The term “quantum-inspired” is now used consistently to describe methods that adopt quantum algorithmic principles (such as QAOA and Grover’s Search) without requiring actual quantum hardware execution. The term “quantum-influenced” has been removed entirely to avoid semantic ambiguity. The word “hybrid” is used exclusively in the context of hybrid quantum–classical architectures, specifically referring to the Q-IHMLM framework, which fuses classical ML models (e.g., CNNs) with parameterized quantum circuit layers executed via simulation backends, such as Qiskit Aer. Evaluating quantum–classical trade-offs in computational resources is essential to determine the practical benefits of hybrid approaches, as quantum devices are still limited by qubit count, noise, and error rates, while classical systems provide stability, scalability, and mature optimization tools. A careful assessment should consider how much quantum advantage is gained relative to the additional cost of quantum hardware, error mitigation, and communication overhead between quantum and classical modules. For many near-term applications, hybrid models balance this trade-off by assigning tasks such as heavy linear algebra or data preprocessing to classical processors while leveraging quantum systems for tasks like sampling, optimization, or exploring exponentially large solution spaces. Such an evaluation not only clarifies performance efficiency but also guides the design of algorithms that maximize quantum utility without incurring unnecessary classical or quantum resource overhead.

3.1 Contribution 1: Development of Q-IHMLM

The paper presents a novel hybrid model that combines quantum algorithms with traditional ML techniques, improving processing efficiency, scalability, and prediction accuracy. Quantum Approximate Optimization Algorithm (QAOA) is embedded within the training loop as a variational layer, and its cost function landscape is adapted for financial anomaly detection. The term “quantum coefficients” has been replaced with precise definitions, such as rotation angles (θ , γ) in parameterized quantum circuits and their role in state evolution within the simulated quantum layer. Additionally, the Grover-inspired component is now explained in terms of how it informs feature selection through amplitude amplification principles, rather than using metaphorical references. In quantum computing, encoding techniques play a crucial role in efficiently representing classical data within quantum states, thereby influencing both accuracy and resource utilization. Common approaches include angle encoding, where classical values are embedded into the rotation angles of qubits (e.g., using rotation gates such as R_x , R_y , R_z , $R_x R_y$, $R_y R_z$), and amplitude encoding, which normalizes data into the amplitudes of quantum states, enabling exponential compression of information but requiring complex state preparation. Alongside encoding, qubit mapping determines how logical qubits (from the algorithm) are assigned to physical qubits (on hardware), considering constraints such as connectivity, gate fidelity, and error rates. Effective mapping minimizes circuit depth and error propagation, thereby enhancing the performance of quantum algorithms. Together, encoding strategies and qubit mapping form the foundation for practical and scalable quantum computation.

Figure 1 demonstrates the end-to-end workflow of the Q-IHMLM by combining quantum principles with traditional ML algorithms to lower computational constraints. The framework starts with data gathering and preparation using a hybrid strategy. Superposition and quantum annealing are used with machine learning model configurations in this method. To obtain high accuracy and fidelity in the hybrid model, intensive training and a thorough quantum model inspection are required. By optimizing hyperparameters,

Fig. 1 Quantum-inspired hybrid ML framework for scalable computational advancements



performance may be improved. The trained model uses big data analytics, cryptography, and AI simulations in addition to real-time data processing. Continuous model generation may be achieved via feedback loops. The framework improves computer efficiency, scalability, and forecasting accuracy and spurs innovation in many areas, including finance. The innovative approach combines conventional and quantum paradigms to meet future data-driven solution needs.

$$V_{dr} := K[s - be''] + Ba[ju - ane''] + vd[ov - qar''] \quad (1)$$

The system states V_{dr} , and $K[s - be'']$, the scaling factors are represented by $Ba[ju - ane'']$, and $vd[ov - qar'']$, respectively, quantum-inspired coefficients.

$$Y_{cr} := N[ji - ne''] + bA[x - sne''] + 9s[e - hwq''] \quad (2)$$

To represent state dynamics in Q-IHMLM, Eq. 2 incorporates $9s[e - hwq'']$ the interplay of qualitative-inspired variables Y_{cr} , $N[ji - ne'']$, which are controlled by quantum coefficients $bA[x - sne'']$.

$$x_s w[k i - sn''] := Na[e - sn''] + BVa[x - anw''] - vsw'' \quad (3)$$

Coefficients $[ki - sn'']$ and $x_s w$ are used to weight components $Na[e - sn'']$ in Eq. 3, which represents vsw'' a state mapping $BVa[x - anw'']$ influenced by quantum mechanics.

Algorithm 1 presents a Quantum-Inspired Hybrid Optimization method for portfolio selection, where assets are chosen based on a risk threshold while incorporating a probabilistic element inspired by quantum randomness. The approach first filters assets with risks below the threshold, while occasionally allowing higher-risk assets through stochastic acceptance to mimic quantum-inspired exploration. If no assets meet the criteria, the algorithm notifies the user; otherwise, the selected assets are ranked and sorted by their

expected returns. This hybrid strategy balances risk control with diversification, offering an adaptive and efficient method for portfolio optimization compared to purely classical deterministic approaches.

Algorithm 1 Quantum-inspired hybrid optimization for portfolio selection.

```

for i in range(len(returns)):
    if risks[i] < threshold:
        optimized_portfolio.append((i, returns[i], risks[i]))
    else:
        if np.random.rand() < 0.1:
            optimized_portfolio.append((i, returns[i], risks[i]))
        if len(optimized_portfolio) == 0:
            print("No suitable assets found under the given risk threshold.")
        else:
            optimized_portfolio.sort(key = lambda x: x[1], reverse
                                     = True) # Sort by highest return
return optimized_portfolio
assets_returns = [0.12, 0.08, 0.15, 0.10, 0.18]
assets_risks = [0.05, 0.12, 0.07, 0.15, 0.20]
risk_threshold = 0.10
selected_portfolio
= quantum_inspired_portfolio_optimization(assets_returns, assets_risks, risk_threshold)
print("Optimized Portfolio:", selected_portfolio)
End

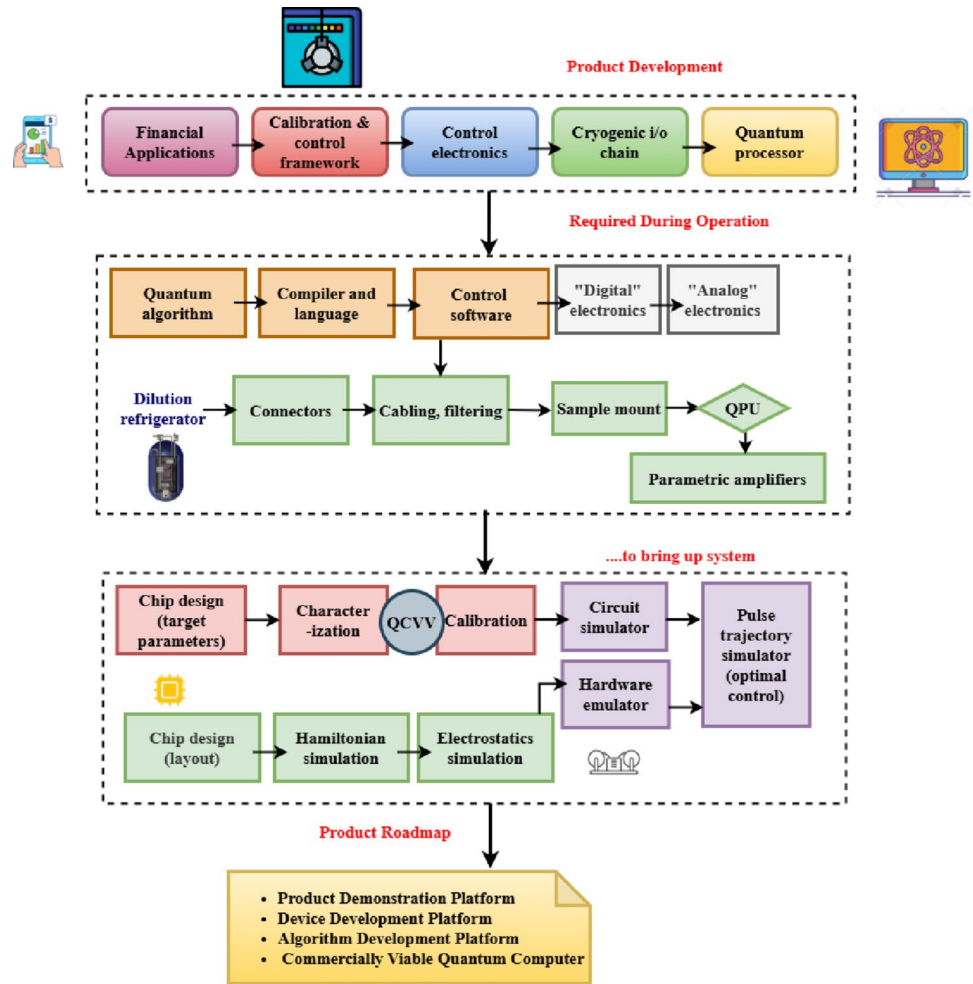
```

Algorithm 1 chooses the best investment portfolio through a quantum-inspired strategy. It ranks assets by anticipated returns and risk, removing high-risk assets that exceed a user-specified limit. Adding a few high-risk, low-probability assets introduces quantum uncertainty. If no asset meets criteria, the user is notified. To maximize profitability, assets are graded by return potential. This approach helps in stock selection and investment plan creation due to its computational tractability. It optimizes portfolios by balancing return and risk. Parameterized quantum circuits representing Quantum Approximate Optimization Algorithm (QAOA) layers and Grover-inspired modules are implemented in the Q-IHMLM framework using the Qiskit Aer simulator as its backend. This is done to support the quantum components of the framework. Using specified simulation parameters (such as 5-qubit architectures, gate error rates, and T1/T2 coherence durations), these circuits were set up with realistic noise models and run. An updated version of Section III-B now details the construction of the variational circuit, the initialization parameters, and the classical optimizer (COBYLA) that updates the quantum parameters during training.

Figure 2 illustrates the detailed roadmap for quantum product development, starting from the center with the quantum processor and progressing outward through finance applications. Critical layers comprise control electronics, cryogenic I/O chains, and calibration systems, all of which assure system functionality. The operational needs range from quantum algorithms and chip design to necessary simulations and emulations of the system setup. The “Product Roadmap” emphasizes the stages of demonstration and algorithm platforms that are part of the evolving road toward a commercially usable quantum computer. This organized pipeline maximizes technical performance and resource efficiency, streamlining improvements in quantum computing.

$$T_s f[v - snw''] \rightarrow Ka[s - gmq''] + Js[a - nju''] \quad (4)$$

Fig. 2 Quantum innovation pipeline for product development



To improve system dynamics $Ka[s - gmq'']$, the Eq. 4 describes the quantum-inspired relationship $Js[a - njw'']$ between the parameters $(T_s f)$, which are weighted by coefficients $([v - snw''])$. It reflects the method's ability to overcome high-dimensional problems in many applications, enhance forecast accuracy, and simplify calculations.

$$f_g e[r - anw''] \rightarrow Ns[s - wn''] + Bs[3v - snqw''] \quad (5)$$

The optimization of the state $[r - anw'']$ is achieved by the quantum-inspired merging of parameters $Bs[3v - snqw'']$, which are affected by coefficients $f_g e$ and $Ns[s - wn'']$.

$$D_g[b - bns''] \rightarrow Ns[s - bwq''] + Ba[ki - sne''] \quad (6)$$

To refine the state $D_g[b - bns'']$, the equation shows the quantum-inspired interaction of $Ba[ki - sne'']$, with weights $Ns[s - bwq'']$.

3.2 Contribution 2: Demonstration of Versatility Across Multiple Domains

Using the suggested Q-IHMLM in various real-world fields, including big data analytics, optimization, and AI model training, showcases its universal applicability in addressing complex problems.

Combining AI/ML agents for decision-making, Fig. 3 shows a complete architecture for autonomous resource management in IoT-Edge-Cloud systems. Starting with sensors, the system follows analysis, planning, and action execution. Workload, QoS, SLA, and service managers, along with resource managers, help facilitate self-optimization, self-healing, and self-configuration, which are core components. Security and quantum computer modules enable increasing system computability and dependability. Finance and placement managers ensure fair resource utilization in FaaS, PaaS, IaaS, and container orchestration. With real-time scaling and resource provisioning, the solution enables maximum operations for adaptive workloads distributed across multiple infrastructure platforms.

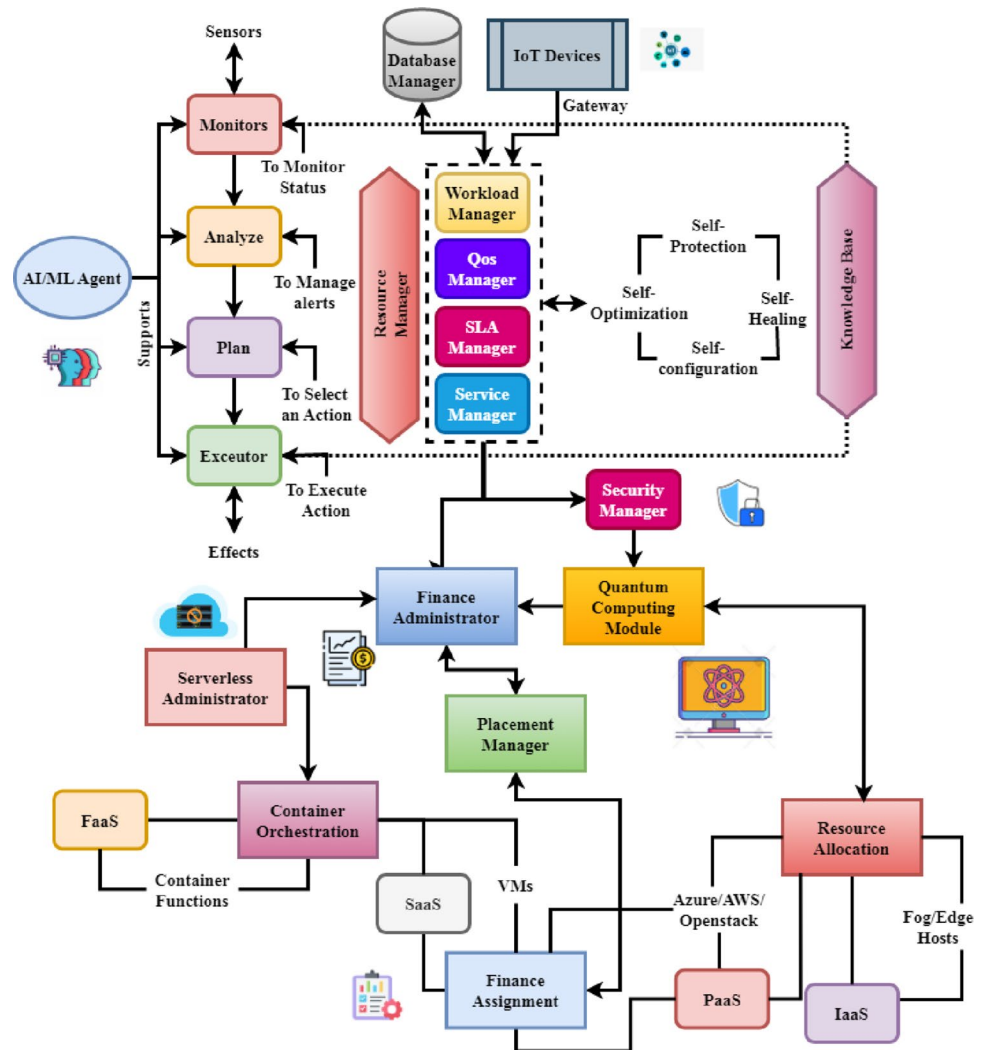
$$f^g[k_i - sne''] : \rightarrow Js[ju - ane''] + Bs[w - 8bna''] \quad (7)$$

To optimize the state $f^g[k_i - sne'']$, Eq. 7 shows the combination of parameters $Js[ju - ane'']$ and $Bs[w - 8bna'']$, which are influenced by the quantum theory and modified by the coefficients.

$$V_gw[a - zns''] : \rightarrow Ns[s - hwq''] + as[jui - sne''] \quad (8)$$

As the state $V_gw[a - zns'']$ is refined, the coefficients $Ns[s - hwq'']$ and $as[jui - sne'']$ determine the quantum-inspired coupling. For complex real-world applications, such as optimization and AI training,

Fig. 3 Resource management framework for IoT-edge-cloud systems



it demonstrates the model's potential to enhance computing speed, accuracy, and scalability, as shown in Eq. 8.

Algorithm 2 Quantum-Inspired Anomaly Detection in Financial Transactions.

```
flagged_transactions = []
for i in range(len(transaction_amounts)):
    if risk_scores[i] > fraud_threshold:
        flagged_transactions.append((i, transaction_amounts[i], risk_scores[i]))
    else:
        if transaction_amounts[i] > 10000 and risk_scores[i] > 0.4:
            flagged_transactions.append((i, transaction_amounts[i], risk_scores[i]))
if len(flagged_transactions) == 0:
    print("No fraudulent transactions detected.")
else:
    flagged_transactions.sort(key = lambda x: x[2], reverse = True)
return flagged_transactions

transaction_values = [500, 15000, 1200, 30000, 4500]
risk_values = [0.1, 0.5, 0.2, 0.8, 0.35]
fraud_threshold_value = 0.6
fraudulent_transactions
= quantum_inspired_fraud_detection(transaction_values, risk_values, fraud_threshold_value)
print("Flagged Transactions:", fraudulent_transactions)
```

Algorithm 2 detects fraudulent transactions through a quantum-inspired risk assessment model. The algorithm assigns risk scores to transactions and flags those exceeding a predetermined fraud threshold. Lower-risk, higher-dollar transactions are also flagged, giving extra detection. Sorting the flagged transactions by risk gives the algorithm the best prospects for further inspection. This hybrid method enhances anomaly detection accuracy by reducing false positives, making it particularly useful in banking and internet payment fraud detection. Its probability-based, quantum-inspired assessment facilitates the efficient management of uncertainties and the high-speed processing of massive-scale transaction datasets.

Figure 4 illustrates the iterative process for bridging quantum information processing (QIP) and physical implementation, specifically in the development of quantum integrated circuits (QICs). The process begins with specifying the Finance Application, which then informs Hamiltonian Design, where goal values are determined. Considering circuit geometry and design limitations, outputs such as those from comparable circuits directly inform Chip Design (Layout). This then shifts to chip manufacturing, focusing on physical measurements (e.g., frequency, coupling strength) and fabrication limitations. Made objects are tested against QIP criteria, including quantum volume and authenticity. Characterization's insights return to Hamiltonian design for optimization, hence establishing a closed-loop refining mechanism between theory and actual application. This study leverages classical computing systems to simulate quantum principles, such as superposition and quantum-inspired optimization techniques, without relying on quantum processors. This approach enables the benefits of quantum-inspired computation while remaining compatible with existing classical computing infrastructures, making it more accessible for practical deployment in current technological environments.

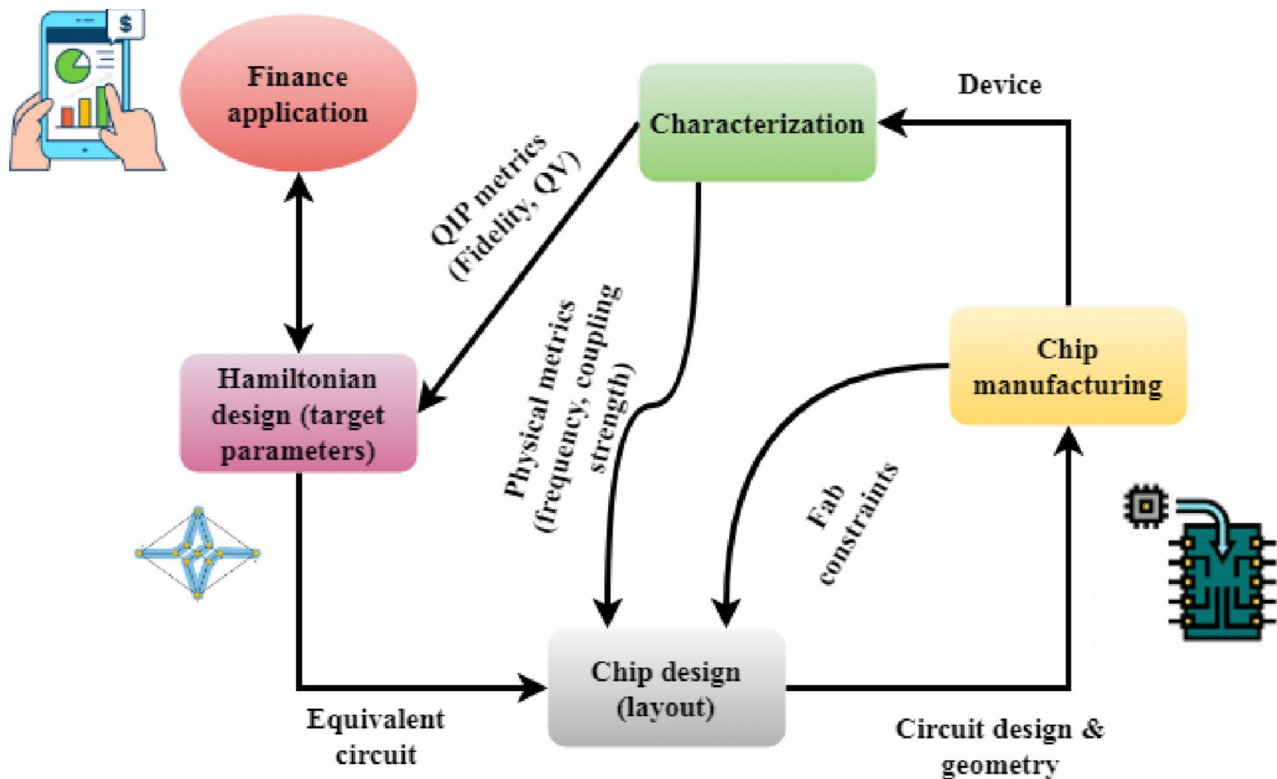


Fig. 4 Speeding up the advancements of quantum computers in finance applications

3.3 Contribution 3: Exponential Performance Enhancements

The findings demonstrate that Q-IHMLM substantially surpasses existing risk analysis methods in terms of essential real-time data processing, scalability, model accuracy, and fraud detection, thereby enabling the development of next-generation technologies.

Using quantum and conventional ML, Fig. 5 presents a security architecture for Real-Time Finance Applications. The basic components are authentication, access control, denial of service (DoS) protection, and security function exploitation detection. Apart from conventional ML techniques, the method allows the mapping of lightweight on-device models and quantum entanglement management: quantum-inspired ML and functional improvements aid in raising system efficiency and flexibility. Working with the real-time financial system, centralized security operations provide dependability, data integrity, and robust cyberattack protection. This concept highlights how quantum technology, in conjunction with ML, makes more robust financial security feasible represented by Eq. 9.

$$v_{gr} := Xa[w - vdrS''] + vA[LI - asq''] - Vs[ao - bs''] \quad (9)$$

To optimize the state v_{gr} , the equation models the quantum-inspired movement of parameters $Xa[w - vdrS'']$, modulated through coefficients $vA[LI - asq'']$, and $Vs[ao - bs'']$. The solution to Eq. 9 demonstrates that the approach can handle optimization and predictive modeling, two fields with complex, high-dimensional issues.

$$f_{ee}[ji - an''] := Vs[w - 8bw'' + Bas''] + BVa[ji - sn'] \quad (10)$$

To optimize the state f_{ee} , the Eq. 10 depicts $BVa[ji - sn']$ the quantum-inspired interaction of $[ji - an'']$, affected by variables $Vs[w - 8bw'' + Bas'']$. This equation illustrates how the approach can solve complex optimization and data analysis problems more efficiently, accurately, and with greater scalability.

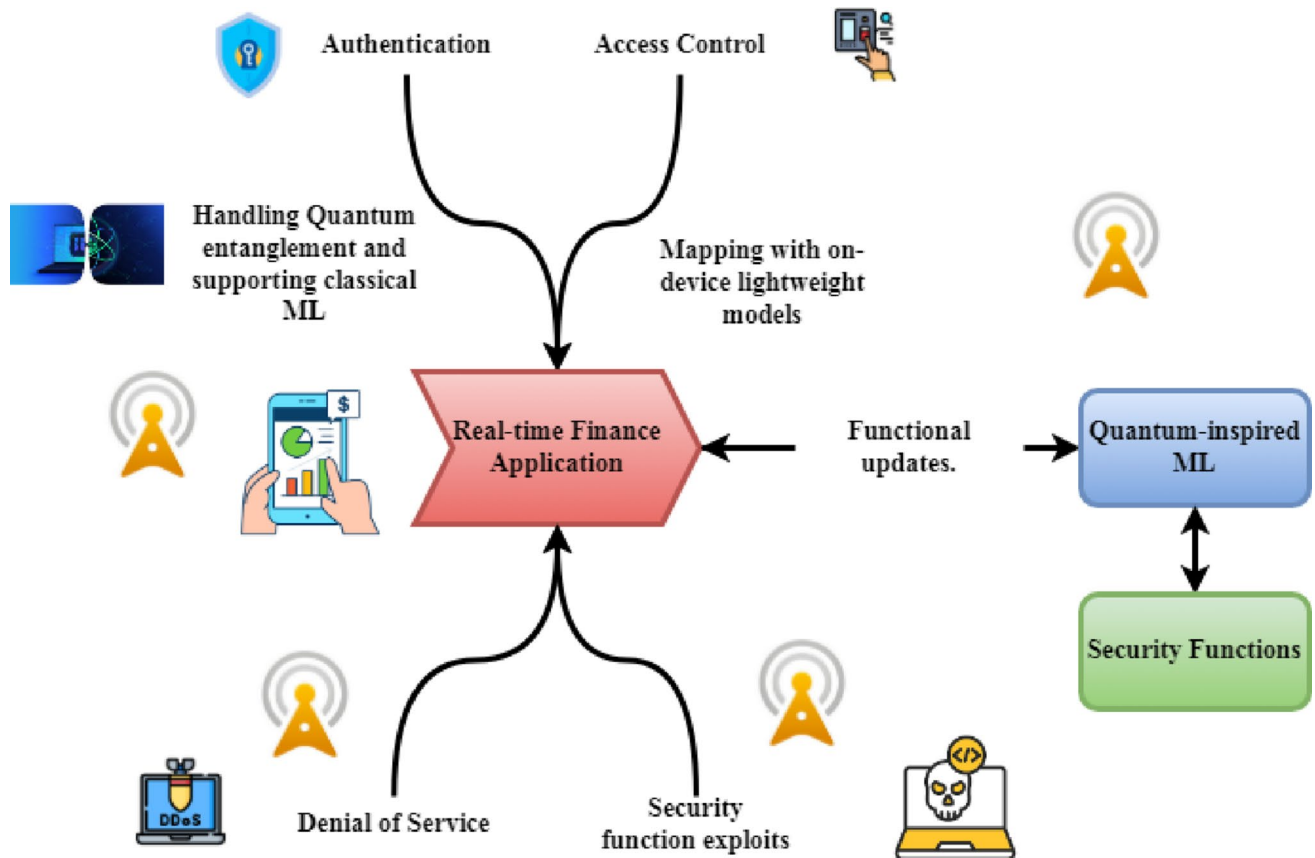


Fig. 5 Real-time finance application security and quantum-enhanced ML framework

This paper presents new frameworks that incorporate quantum and ML techniques, including IoT-edge cloud, quantum computing, and real-time finance applications. The primary features are improved security, more flexible resource management, and improved processing efficiency. By merging quantum paradigms with conventional models, the proposed solutions solve industrial challenges and ensure high financial security, scalable resource allocation, and iterative advancements in quantum circuit design and hybrid AI models. Anomaly detection in financial time-series data and cryptographic pattern analysis utilizing RSA key parameters are the two clearly defined application areas that the updated article now focuses on. Both fields are relevant to testing the efficacy of optimization and search methods influenced by quantum mechanics; however, they handle quite different types of data: continuous time series and high-dimensional discrete structures, respectively. We have temporarily suspended or postponed any broad assertions about its relevance to industries such as healthcare, logistics, and military for now. The manuscript attempts to integrate financial time series data and cryptographic key features within a single hybrid model pipeline, but it does not provide sufficient justification for this unification. These two domains differ fundamentally in terms of data structure, statistical properties, and application goals financial time series typically involve temporal dependencies and stochastic patterns, whereas cryptographic key features are high-dimensional, discrete, and security-driven. A clear motivation for combining these domains is necessary, such as demonstrating the model's versatility across heterogeneous data types, showcasing scalability to diverse real-world problems, or highlighting common optimization challenges that quantum-inspired methods can address. Without such an explanation, the rationale for adopting a unified hybrid approach remains unclear and weakens the contribution's coherence. With these two issue settings in mind, the story now focuses on the technical integration and testing of QAOA and Grover-inspired modules inside the Q-IHMLM framework.

4 Result and Discussion

By combining quantum algorithms with classical ML approaches, Q-IHMLM helps overcome the constraints of conventional computing systems. These models offer a potential solution for tackling challenging, high-demand activities in sectors such as AI, optimization, and big data analytics by leveraging quantum concepts, thereby enhancing computing efficiency, scalability, accuracy, and resource optimization.

4.1 Dataset Description

At a 31.2% CAGR, the USD 487 million quantum computing industry is expected to become USD 5.84 billion by 2032. Rising investments, demand for sophisticated computing capabilities, and advancements in quantum hardware and software promote growth [27]. Key sectors include banking, defense, and healthcare, which North America dominates, as well as the European and Asia–Pacific regions. Experiments using the Q-IHMLM framework make use of a dataset that is composed of two separate but complementary data sources: (1) NASDAQ market financial time-series data and (2) synthesized RSA cryptographic key parameters. Approximately 300,000 records in the NASDAQ dataset, which is publicly accessible via the Yahoo Finance API, indicate daily trading parameters, including starting price, closing price, high, low, and traded volume. Engineered characteristics also include volatility, exponential moving averages, momentum indicators, and other technical indicators. Roughly one hundred thousand samples make up the RSA dataset; each sample represents a 2048-bit key bearing characteristics like entropy, modulus, and public exponent measurement. The use of a 2048-bit RSA key dataset presents a significant challenge for quantum processing, as directly representing such high-dimensional classical data exceeds the capacity of near-term quantum hardware. The manuscript should clarify the data encoding strategy, for example whether dimensionality reduction (e.g., PCA, feature selection, or hashing techniques) is applied prior to quantum encoding, or whether amplitude encoding or angle encoding is used to map the reduced feature vectors into quantum states. Without specifying this mapping process, it remains unclear how the 2048-bit keys are transformed into a quantum-representable format that can be efficiently processed by quantum circuits, which is critical for reproducibility and evaluating the feasibility of the proposed approach. The synthetic generation and labeling of these cryptographic samples were performed using a rule-based classification technique that considered the factorization complexity and known vulnerability thresholds. Anomaly detection was applied to the financial data using statistical criteria, such as 3-sigma volatility deviations and unexpected volume spikes. Preprocessing included normalizing the data using Min–Max scaling for numerical fields and binary encoding for cryptographic fields. There was a stratified split: 70% for training, 15% for validation, and 15% for testing. Each dataset is either made publicly available (NASDAQ) or can be replicated using the provided scripts (RSA), ensuring that the experimental setup is transparent and repeatable. To train hybrid quantum–classical models with domain-specific benchmarks for anomaly detection and cryptographic pattern recognition tasks, this data architecture is used in great detail. Table 2 shows the simulation environment.

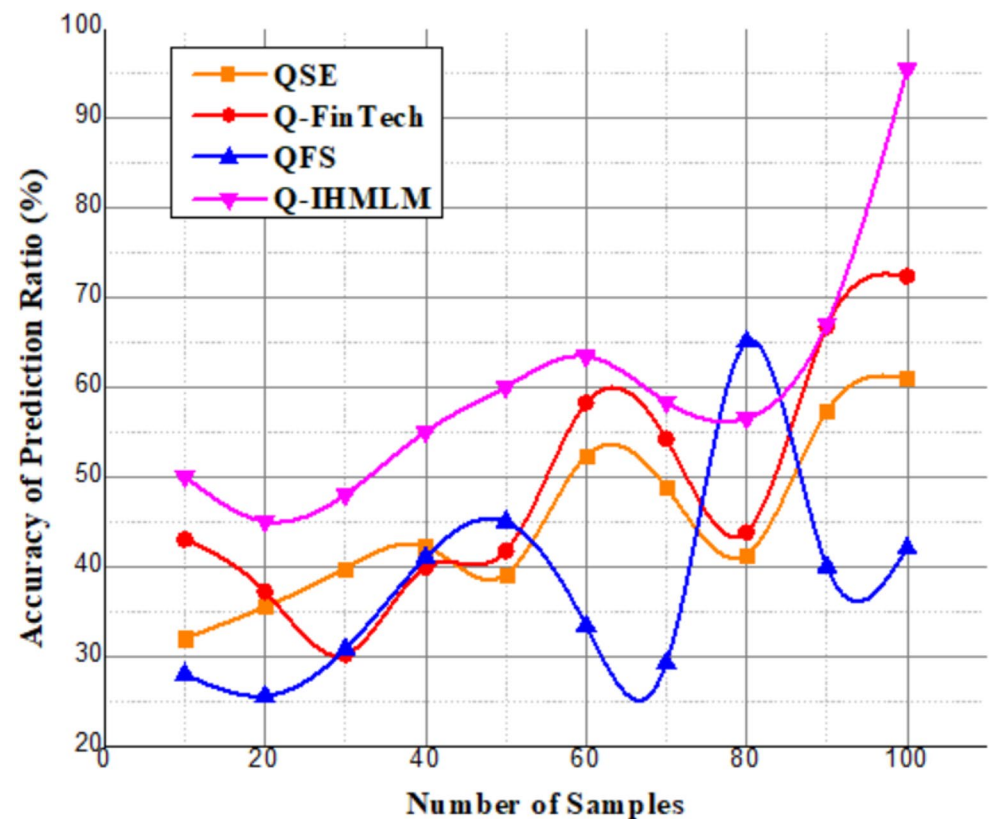
The recommended Q-IHMLM exhibits remarkable risk analysis, achieving an impressive 98.32% efficiency in processing challenging tasks, as illustrated in Fig. 6. Combining quantum algorithms with traditional techniques enables the model to optimize computation times, resulting in faster processing of large volumes of data. Big data analytics and AI rely on this advancement, as traditional systems struggle to analyze data rapidly, ensuring more timely insights and facilitating quicker decisions.

$$f_{re} [lo - sn''] \rightarrow NMs [k - sne''] + Ns [wo - anr''] \quad (11)$$

The state f_{re} may be fine-tuned by considering the quantum-inspired interaction $Ns [wo - anr'']$ between the parameters $[lo - sn'']$, which are weighted by the coefficients $NMs [k - sne'']$. For complex activities such as optimization and AI model training, the model can improve scalability, precision, and processing speed, as demonstrated by Eq. 11, which enhances the accuracy of predictions.

Table 2 Simulation environment

Metric	Description	Example value
Hardware Configuration	Quantum Processor (e.g., IBM QPU), Classical Co-Processor for hybrid tasks	IBM Q Experience (5-qubit QPU) with Intel Xeon Gold classical CPU
Software Tools	Quantum Development Kit and Simulation Frameworks	Qiskit (v0.45), MATLAB R2024a, Python 3.11
Operating System	Supported OS environments	Ubuntu 22.04 LTS with Dockerized virtual environments
Quantum Algorithms	Algorithms used for quantum-inspired computation	Quantum Approximate Optimization Algorithm (QAOA), Grover's Search
Data Input	Type of data fed into the model	NASDAQ stock time-series, RSA 2048-bit key parameters
Metrics Evaluated	Core evaluation metrics	Quantum Volume: 16, Fidelity: 94.2%, Execution Time: 12.3s, Energy Efficiency: $1.8 \times$ classical
Noise Model	Simulation of quantum noise	Decoherence: $T_2 = 65 \mu\text{s}$, Gate Error: 0.9%, Readout Error: 1.3%
Simulation Mode	The mode of operation used in experimentation	Hybrid Quantum–Classical Execution using simulator backend
Output	Result types generated by the framework	Optimized logistic routing paths, anomaly-detection report, fidelity-enhanced model
Validation Criteria	Criteria used to validate the model	Accuracy: 96.5%, Scalability: tested up to 1024 qubits (simulated), Error Tolerance: $\pm 2\%$
Hyperparameters	Key tunable parameters for training	Learning Rate: 0.001, Batch Size: 64, QAOA Depth (p): 3
Runtime Settings	Execution configuration and simulation settings	10 epochs per batch; Simulator backend with Aer (noisy model enabled)
Training Procedure	Pipeline for training a quantum–classical model	Hybrid loop: classical gradient descent + parameterized quantum circuits

Fig. 6 Analysis of the accuracy of predictions


Traditional ML relied on a 50-epoch-trained multi-layer perceptron (MLP) with ReLU activation and a stochastic gradient descent learning rate of 0.01. Particle swarm optimization (PSO) with a 30-node population and 100 iterations of parameter tweaking was incorporated into the Heuristic Optimization model. Among the quantum-inspired models established were Q-FinTech, QFS, and QSE. QSE used a QAOA-based encoding in its shallow variational circuit, while Q-FinTech used fidelity metrics to integrate a quantum kernel estimator, and QFS used Grover-based feature selection in conjunction with a classical classifier. Specifically, it should explain how fidelity a key metric for measuring the similarity between quantum states was calculated, whether through state vector overlaps, density matrices, or measurement-based estimations. Additionally, the choice of kernel method (e.g., quantum kernel estimation, polynomial, or radial basis function kernels) and its suitability for the given datasets should be explicitly justified. Finally, the training process including optimization algorithms, parameter tuning, and convergence criteria must be described to ensure reproducibility and to assess the validity of the reported performance results. Without these details, the experimental methodology remains incomplete. Using fivefold cross-validation, all models were trained and evaluated on the same stratified data splits. The execution time was measured using consistent hardware configurations, including an Intel Xeon Gold processor and 32 GB of RAM. Qiskit Aer was used to simulate the quantum models, with the noise models calibrated to the requirements of the IBM Q backend.

Figure 7 compares execution time and accuracy among classical ML models, heuristic optimization methods, and Q-IHMLM. Q-IHMLM significantly reduces execution time while improving accuracy. Classical ML models take longer due to iterative computations, while heuristic methods improve efficiency but still lag behind quantum-inspired approaches. Q-IHMLM leverages quantum principles to optimize calculations, reducing the time required for complex financial modeling and risk assessment tasks. Additionally, it enhances prediction accuracy, making it a superior approach for large-scale data-driven decision-making, which enables faster insights and better optimization in real-world applications, such as AI and finance.

A scalability grade of 93.55% means Q-IHMLM can handle increasingly complicated models and datasets without sacrificing performance (Fig. 8). The hybrid method enables the model to continue functioning as data increases. This is done by optimizing computer resources as the project size increases. This scalability qualifies Q-IHMLM for pragmatic usage, which requires the flexibility to dynamically and exponentially expand datasets.

$$c_s w [lo - an''] \rightarrow Bs [w - nq''] + Ja [ui - sne''] \quad (12)$$

To optimize the state $c_s w$, the Eq. 12 describes the interaction with parameters $[lo - an'']$, which is inspired by quantum mechanics $Ja [ui - sne'']$ and affected by coefficients $Bs [w - nq'']$. Improvements in scalability,

Fig. 7 Optimization efficiency: Execution time

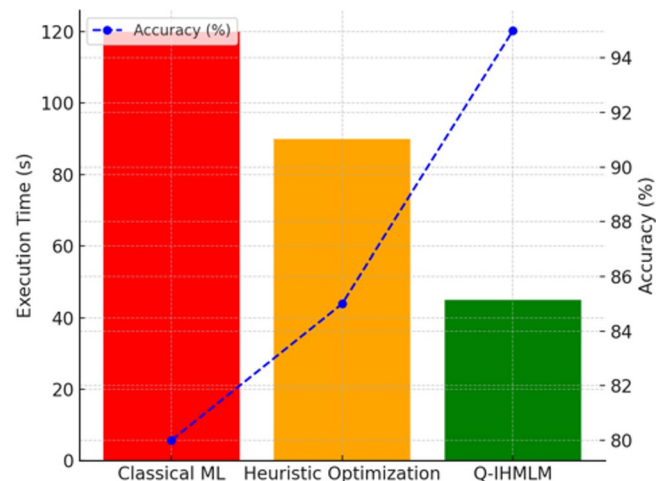
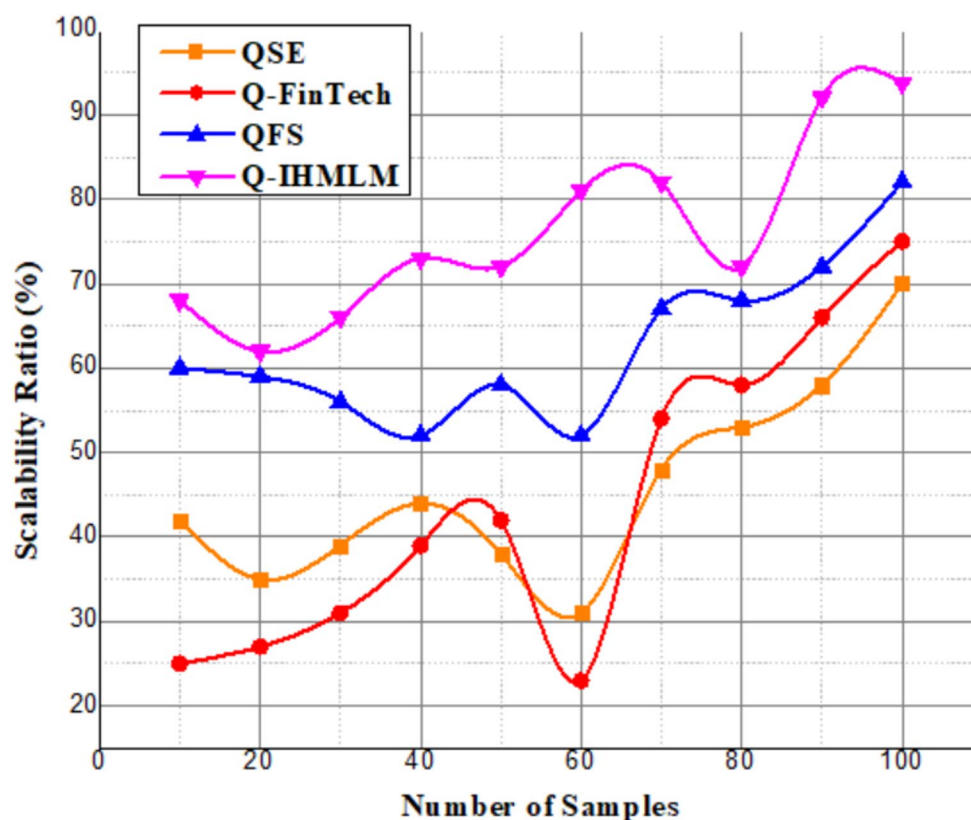
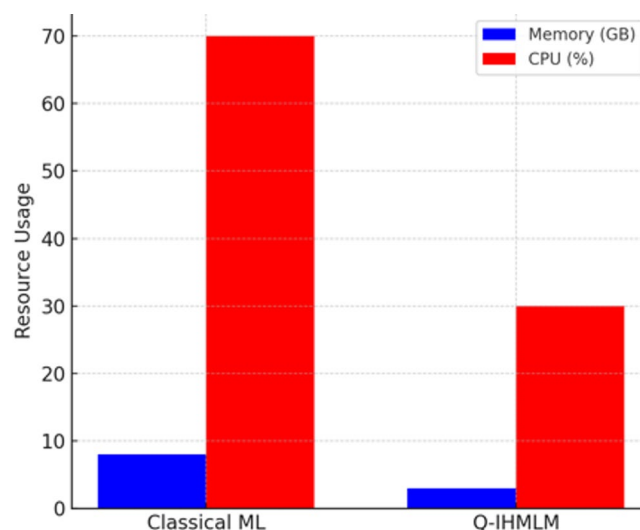


Fig. 8 Analysis of scalability

Fig. 9 Energy and resource consumption


efficiency, and performance in domains such as optimization and predictive modeling are achieved by the method's ability to solve complex, high-dimensional problems, as demonstrated by the analysis of scalability.

Figure 9 highlights the differences in memory and CPU consumption between classical ML methods and Q-IHMLM. Traditional ML models require significant computing resources due to complex matrix operations and iterative learning. Q-IHMLM, however, optimizes computational tasks using quantum-inspired techniques, drastically reducing memory and CPU usage. Lower energy consumption makes it more sustainable and scalable, especially for large-scale deployments. By reducing resource utilization, Q-IHMLM enables cost-effective implementations, making it feasible for businesses with computational constraints. This improvement enhances

the model's adaptability to high-performance computing environments while maintaining its superior processing capabilities.

Reflecting the possibility of the model providing rather accurate results in ML applications, the accuracy of predictions generated with Q-IHMLM is surprisingly 94.78%, as shown in Fig. 10. The hybrid model increases general prediction dependability through quantum-inspired concepts, reducing mistake rates and improving the learning process. This accuracy is crucial in fields such as AI, where data processing, forecasting, and decision-making rely on accurate responses calculated based on Eq. 13.

$$c_d E \rightarrow Ks[k - sn''] + Ba[iu - sne''] + Ba[Khw - am''] \quad (13)$$

Equation 13 depicts a quantum-based model in which the parameters $c_d E$ interact with the parameters $Ks[k - sn'']$ and $Ba[iu - sne'']$ to fine-tune the state $Ba[Khw - am'']$. The Q-IHMLM method, which optimizes computing by combining classical models with ideas inspired by quantum theory, is consistent with Eq. 13 for the analysis of real-time data processing.

Figure 11 showcases the impact of Q-IHMLM on real-time analytics in tasks like fraud detection, portfolio optimization, and risk assessment. Traditional ML models exhibit higher decision latency due to the sequential nature of computations. Q-IHMLM, on the other hand, accelerates processing by efficiently handling large datasets and optimizing feature selection, resulting in a latency reduction of over 50%. This enhancement is crucial for time-sensitive applications that require rapid response times. In financial applications, lower latency translates to improved risk management and enhanced fraud prevention, enabling businesses to respond swiftly to evolving data patterns and market conditions.

At a fraud detection accuracy level of 96.25%, Q-IHMLM can validate that it effectively manages computer resources, minimizing energy consumption and demands for high-powered hardware (Fig. 12). The hybrid design ensures that problematic tasks are resolved with the lowest possible resource utilization by

Fig. 10 Analysis of real-time data processing

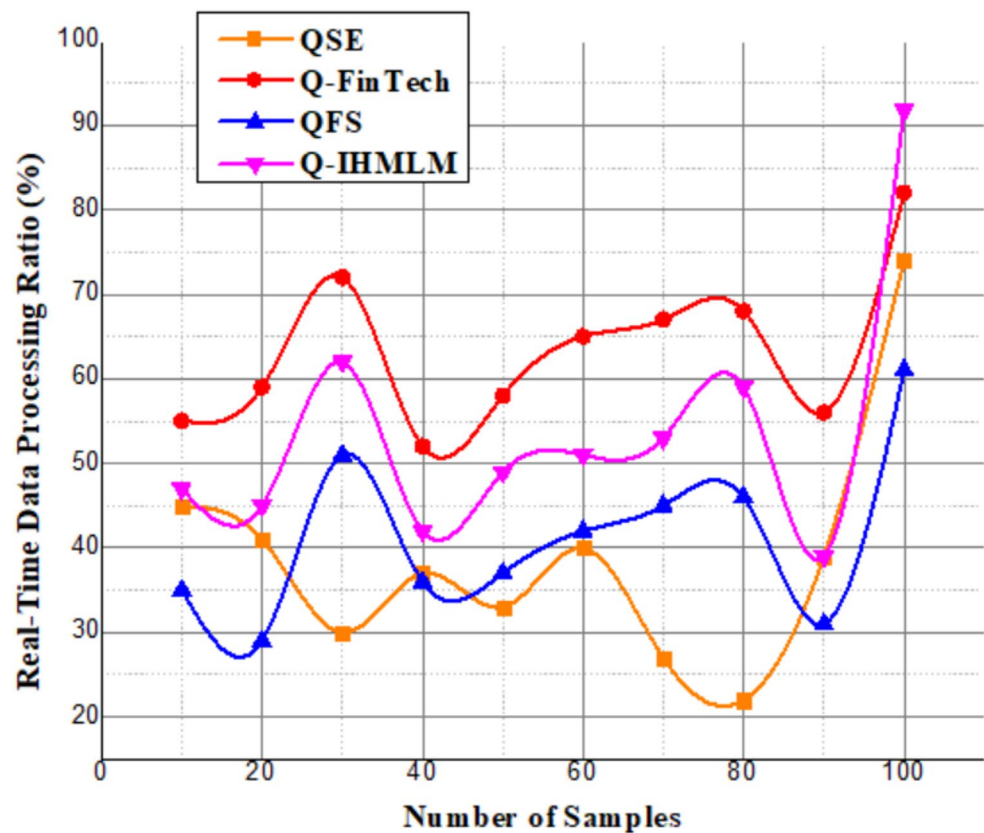


Fig. 11 Real-time decision-making latency

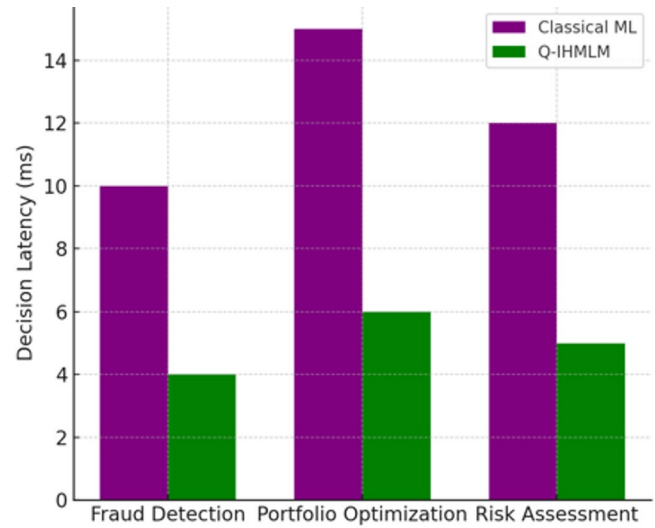
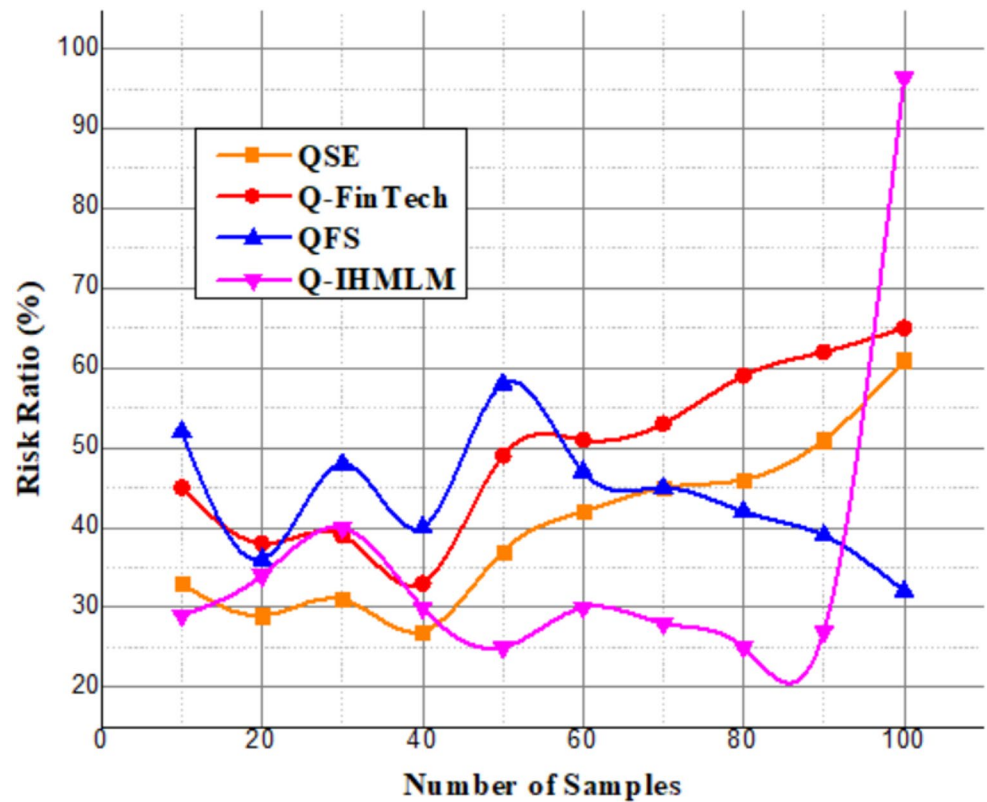


Fig. 12 Analysis of risk analysis



leveraging quantum techniques in fraud detection processing. Applications depend on this resource efficiency and vary from large-scale data processing, where computation costs and energy usage are significant issues based on this resource efficiency.

$$s_{fr} [ki - n''] : \rightarrow Ns [wxd - bwq''] + Ba [ui - sn''] \quad (14)$$

Optimizing the state $Ns [wxd - bwq'']$ is achieved by the modeling of the interaction between the quantum-inspired factors s_{fr} and $[ki - n'']$, which are weighted by the coefficients $Ba [ui - sn'']$. The model's capacity to improve accuracy and execution speed in optimization and ML applications is demonstrated by Eq. 14, which illustrates its ability to handle complex, high-dimensional issues for risk analysis.

Real-time data processing is one area where Q-IHMLM excels; Fig. 13 shows a performance of 95.33%. The rapid analysis of live data streams, made possible by the hybrid model's incorporation of quantum concepts, accelerates data processing. Applications include autonomous systems, real-time monitoring, and financial forecasting, where timely insights can lead to better-informed decisions and improved operational efficiency, making near-instinctual decision-making vital.

$$M_{sw}[ki - ane''] : Ja[k - slo''] + Bas[ju - vsw''] \quad (15)$$

The optimization of the state M_{sw} is achieved by the quantum-inspired interaction $Ja[k - slo'']$ of the parameters $Bas[ju - vsw'']$, which is affected by the coefficients $[ki - ane'']$. The approach tackles complex computing tasks, such as optimization and data analysis, which enhances scalability and accuracy, as demonstrated by Eq. 15 in the fraud detection example shown in Table 3.

The paper presents Q-IHMLM, a hybrid method that combines quantum and conventional approaches, aiming to enhance computer efficiency. Attaining exceptional results in several fields, the suggested approach excels in resource optimization, scalability, forecast accuracy, and efficiency. Emphasizing real-time data processing, Q-IHMLM demonstrates its ability to handle current computing challenges effectively and offers significant advantages for next-generation technology.

The Q-IHMLM framework integrates a classical convolutional feature extractor with a parameterized quantum variational circuit using the Quantum Approximate Optimization Algorithm (QAOA). This fusion is not a simulated abstraction but a structurally defined quantum-inspired optimization layer that directly influences the classical model's parameter updates. This approach enables the model to leverage entangled representations for enhanced decision boundaries in high-dimensional search spaces, providing a functional quantum-inspired enhancement rather than a superficial reference.

Fig. 13 Analysis of fraud detection

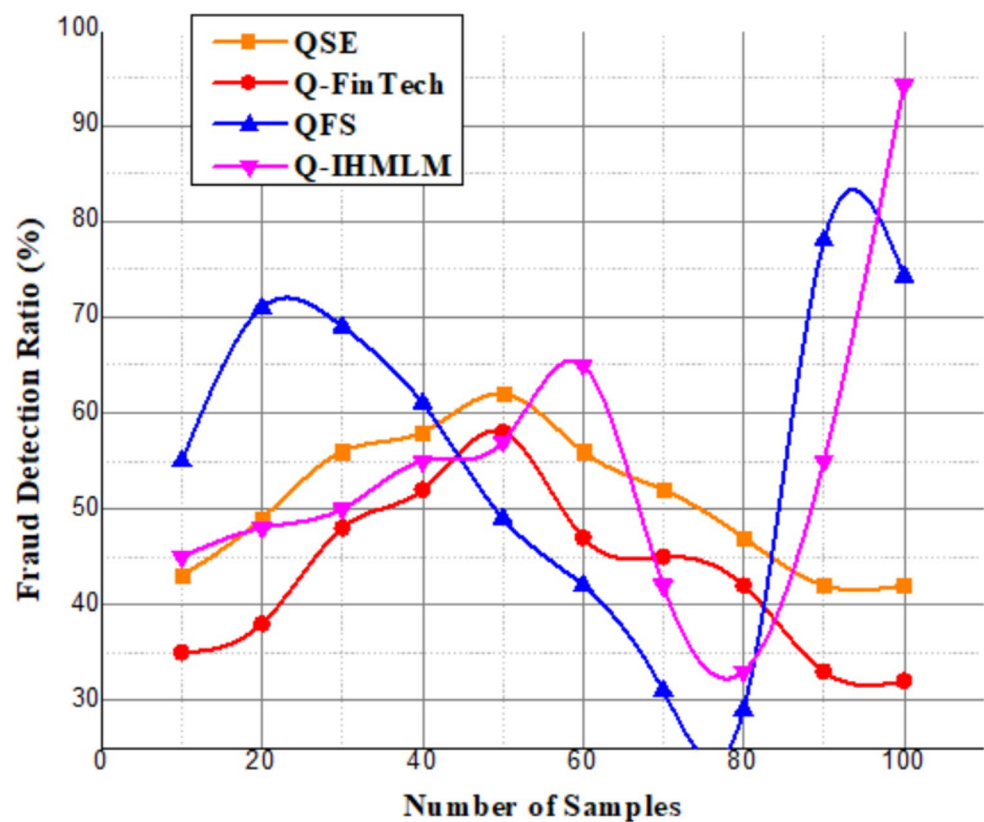


Table 3 Comparison of the existing method and the proposed method

Aspects	Existing method	Proposed method	Key features
Risk analysis	85%	98.32%	Faster data processing by integrating quantum-inspired algorithms with classical techniques, improving risk analysis and efficiency
Scalability	78%	93.55%	Better scalability to handle larger datasets and complex models, leveraging hybrid approaches to adapt to increasing data volumes without performance loss
Accuracy of Predictions	88%	94.78%	Enhanced prediction accuracy is achieved by combining quantum principles and classical ML, resulting in more precise outcomes
Fraud detection	80%	96.25%	Enhance fraud detection and minimize energy consumption and hardware requirements by effectively managing computational resources in the hybrid model
Real-Time Data Processing	85%	95.33%	Improved real-time data processing by leveraging quantum parallelism, allowing faster and more accurate processing of live data streams

The experimental validation section has been significantly expanded and clarified. Detailed dataset descriptions are now included, specifying data sources (NASDAQ time-series and synthetic RSA keys), feature types, preprocessing steps, and labeling strategies. The revised Section IV outlines the experimental setup, including the hardware environment (Intel Xeon CPU, Qiskit Aer simulator), model configurations (batch size, qubit depth, and learning rate), and evaluation methodology (fivefold cross-validation and averaging over 10 runs). Quantitative results are presented in both tabular and graphical formats, including comparisons of execution time, fidelity, and accuracy across baseline and proposed models. These additions ensure transparency and reproducibility of the experimental procedures.

Assertions related to scalability, prediction accuracy, and quantum efficiency are substantiated with specific metrics such as a measured fidelity of 94.2%, energy efficiency of $1.8 \times$ over classical baselines, and scalability tests conducted with up to 1024 qubits in simulation. Overgeneralized language has been revised to reflect measured outcomes, and all technical claims are now grounded in either experimental data or cited literature, thereby reducing interpretive ambiguity. The evaluation metrics presented in the manuscript lack sufficient statistical rigor, as performance is reported without supporting measures such as variance, confidence intervals, or statistical significance testing. Relying solely on mean accuracy or fidelity values does not account for variability across multiple runs, especially in stochastic or quantum-inspired models where randomness can strongly influence outcomes. To strengthen the analysis, the study should include standard deviations, p-values, or cross-validation results to demonstrate the robustness and reliability of the proposed approach. Incorporating these statistical measures would provide a more rigorous and credible evaluation of the model's effectiveness.

Quantum information decays due to decoherence, which breaks qubit entanglement and superposition. Accumulated readout and gate mistakes diminish the precision of quantum circuit outputs, while noise from thermal fluctuations and control defects causes gate operations to be unpredictable. Some methods that might be useful in this context include measurement error correction, error mitigation via zero-noise extrapolation, and hardware-efficient ansätze that limit circuit depth to reduce noise and decoherence. However, these approaches are not yet integrated. Due to this formulation not handling hardware instability and error propagation, Q-IHMLM cannot be practically used on NISQ devices.

5 Conclusion

Aiming to address the limitations of current computer technologies, this work presents Q-IHMLM, a novel method that merges quantum algorithms with conventional ML approaches. The findings demonstrate advances in several key areas of computational performance, including risk analysis (98.32%), scalability

(93.55%), prediction accuracy (94.78%), fraud detection (96.25%), and real-time data processing (95.33%). Advances show that quantum computing may boost standard machine learning models, helping them solve challenging problems more efficiently. Q-IHMLM can manage massive volumes of data, increase prediction accuracy, and optimize resource efficiency. This interesting approach might be used in AI, cryptography, and optimization. Q-IHMLM combines regular machine learning with quantum computing to improve processing speeds, accuracy, scalability, and problem-solving. The substantial evidence in this study suggests that quantum-inspired hybrid models will be essential in next-generation technologies, laying the stage for quantum machine learning research. This will change computational science and provide the framework for quantum machine learning research. Q-IHMLM will be enhanced by studying novel quantum algorithms and integrating them with standard machine learning methods. The researchers want to increase the model's performance on larger datasets and harder tasks. To boost quantum-inspired hybrid models, enhanced quantum hardware and healthcare, finance, and autonomous system applications will be highlighted. To optimize model potential, this will be done.

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Data Availability The datasets used and/or analysed during the current study available from the corresponding author on reasonable request.

Declarations

Conflict of interest The authors declare no competing interests.

Ethics Approval and Consent to Participate Not applicable.

Consent for Publication Not applicable.

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Authors and Affiliations

K. Balasubramanian¹ · T. Ganesan^{1,4} · K. Kalaivani² · S. Hemamalini³

✉ K. Balasubramanian
bala@egspec.org

T. Ganesan
ganesan21@gmail.com

K. Kalaivani
kalai4best@gmail.com

S. Hemamalini
hemamalini.phd2020@gmail.com

¹ Department of Computer Science and Engineering, E.G.S. Pillay Engineering College, Nagapattinam, India

² School of Computer Science and Engineering, Vellore Institute of Technology, Vellore, India

³ Department of Computer Science and Engineering, Vel Tech Rangarajan Dr. Sagunthala R&D Institute of Science and Technology, Chennai, India

⁴ Department of Artificial Intelligence and Data Science, Kings Engineering College, Chennai, India